Engine Model(s): PT6A

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ENGINE, TURBOPROP - DESCRIPTION AND OPERATION

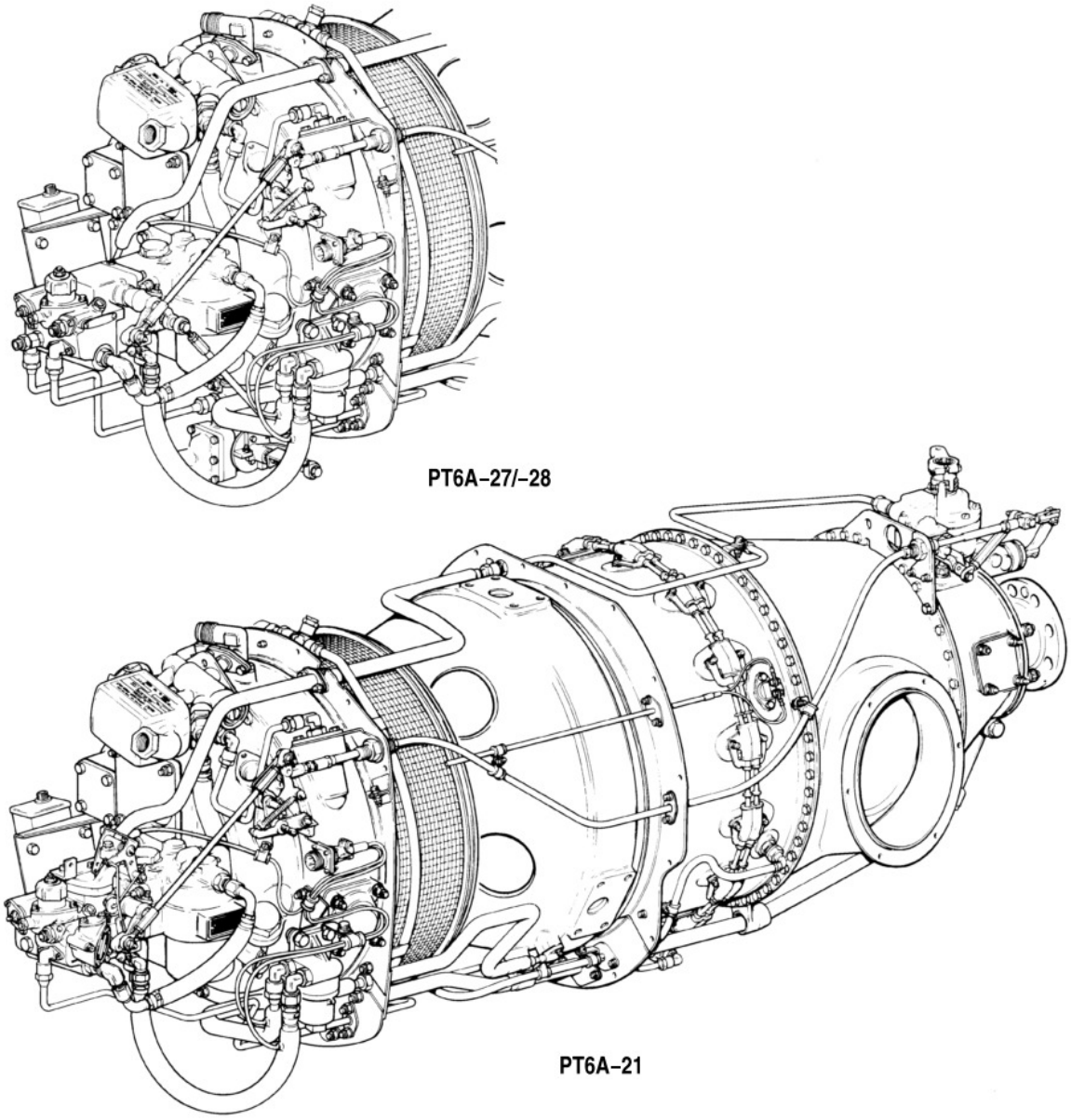
1. Description and Operation

The PT6A Series power plant is a lightweight free turbine engine. The engine utilizes two independent turbine sections: one driving the compressor in the gas generator section and the second driving the propeller shaft through a reduction gearbox. The engine is self-sufficient since its gas generator driven oil system provides lubrication for all areas of the engine, pressure for the torquemeter and power for propeller pitch control.

Refer to Figs. 1 and 2 for the main features of a typical engine, Fig. 3 for the engine cross-section and Fig. 4 for engine stations, flanges and bearings.

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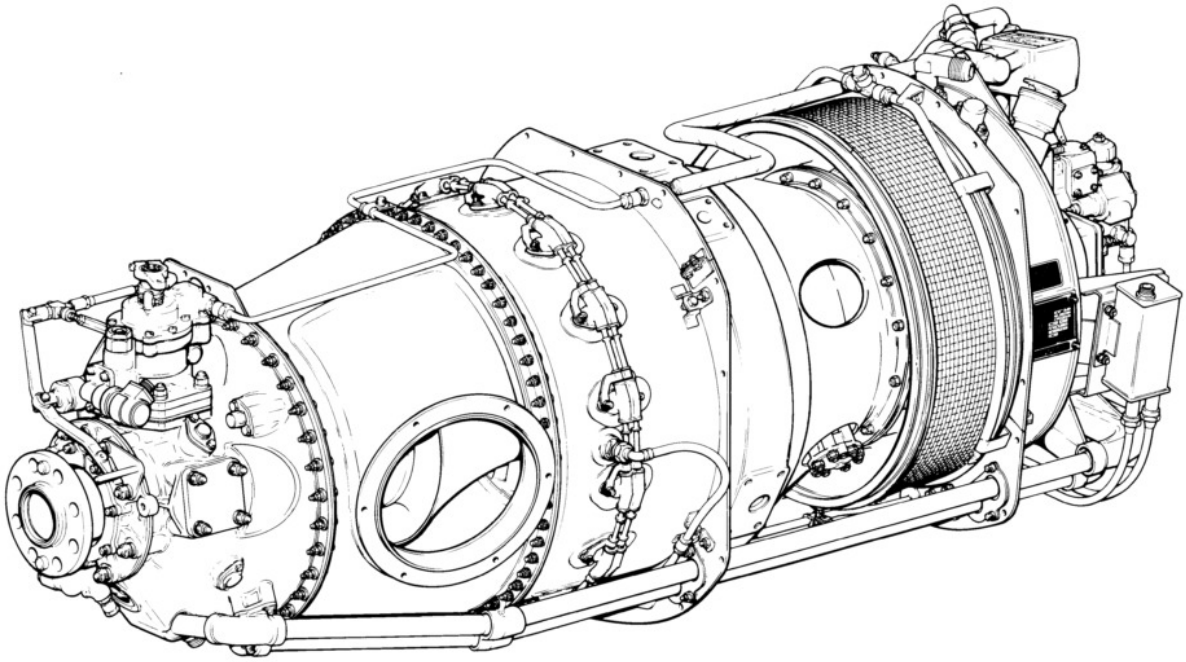
Figure 1 Right Rear View of Engine



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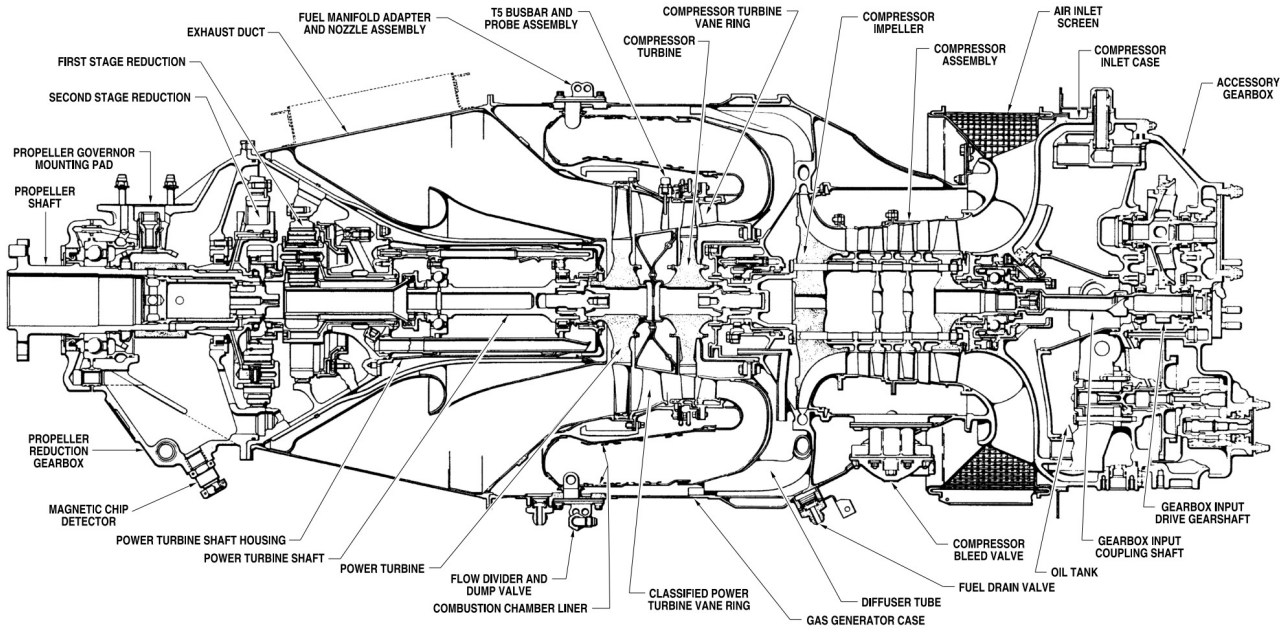
Figure 2 Left Front View of Engine



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Figure 3 Engine Cross Section

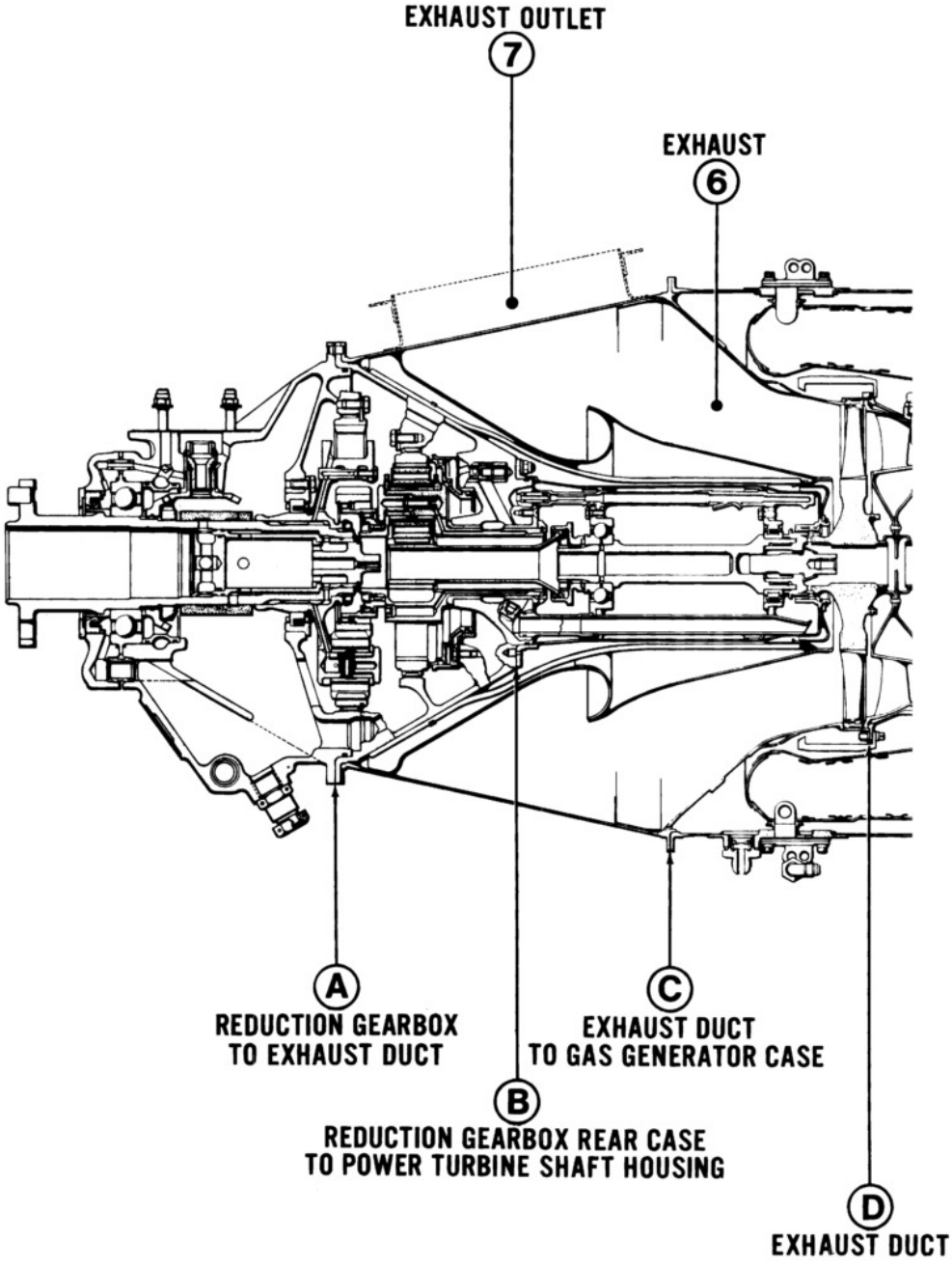


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Figure 4 Engine Stations and Flanges (Typical)

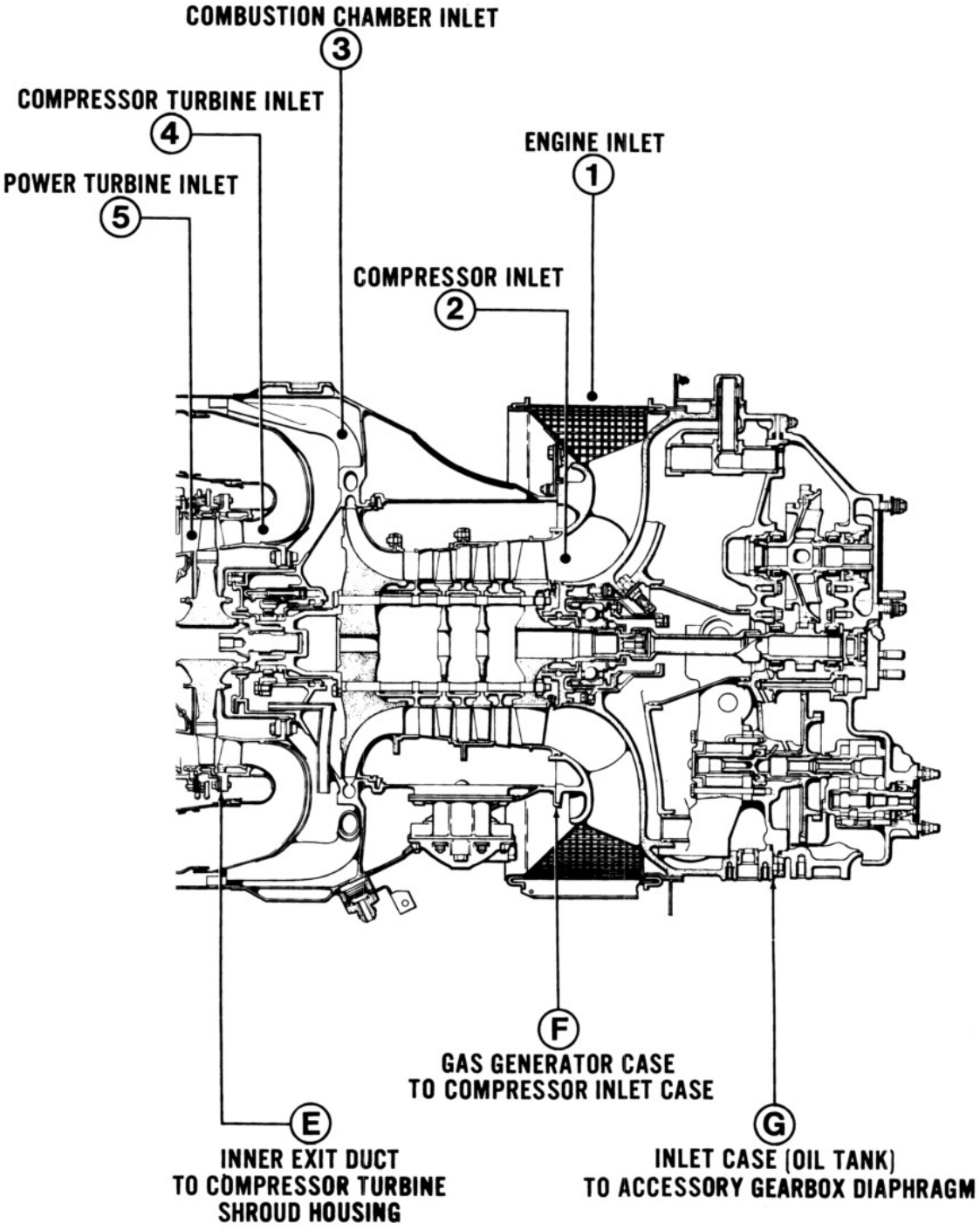
(SHEET 1 OF 2)



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(SHEET 2 OF 2)



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The inlet air enters the engine through an annular plenum chamber, formed by the compressor inlet case, where it is directed forward to the compressor. The compressor consists of three axial stages combined with a single centrifugal stage, assembled as an integral unit.

A row of stator vanes, located between each stage of compression, diffuses the air, raises its static pressure and directs it to the next stage of compression. The compressed air passes through diffuser tubes which turn the air through ninety degrees in direction and converts velocity to static pressure. The diffused air then passes through straightening vanes to the annulus surrounding the combustion chamber liner and the gas generator case.

The combustion chamber liner consists of an annular weldment having perforations of various sizes that allow entry of compressor delivery air. The flow of air changes direction 180 degrees as it enters and mixes with fuel. The fuel/air mixture is ignited and the resultant expanding gases are directed to the turbines. The location of the liner eliminates the need for a long shaft between the compressor and the compressor turbine, thus reducing the overall length and weight of the engine.

Fuel is injected into the combustion chamber liner through 14 simplex nozzles arranged for ease of starting. Fuel is supplied by a dual manifold consisting of primary and secondary transfer tubes and adapters. The fuel/air mixture is ignited by two spark igniters which protrude into the liner. The resultant gases expand from the liner, reverse direction in the exit duct zone and pass through the compressor turbine inlet guide vanes to the compressor turbine. The guide vanes ensure that the expanding gases impinge on the turbine blades at the correct angle, with minimum loss of energy. The still expanding gases are then directed forward to drive the power turbine.

The compressor and power turbines are located in the approximate center of the engine with their respective shafts extending in opposite directions. This feature provides for simplified installation and inspection procedures. The exhaust gas from the power turbine is directed through an annular exhaust plenum to atmosphere via twin opposed exhaust ports provided in the exhaust duct.

Interturbine temperature (T5) is monitored by a cold junction thermocouple system comprising a bus-bar, probes and harness assembly installed between the compressor and power turbines with the probes projecting into the gas path. A terminal block mounted in the gas generator case provides a connection point to cockpit instrumentation and to a T5 trim thermocouple mounted externally in the air inlet zone.

All engine-driven accessories, with the exception of the propeller governor, Nf overspeed governor and Nf tachometer-generator are mounted on the accessory gearbox at the rear of the engine. These components are driven by the compressor by means of a coupling shaft which extends the drive through a conical tube in the center section of the oil tank. The rear location of accessories provides for a clean engine and simplifies maintenance procedures.

The engine oil supply is contained in an integral oil tank which forms the rear section of the compressor inlet case. The tank has a total capacity of 2.3 U.S. gallons and is provided with a dipstick.

Fuel supplied to the engine from an external source is further pressurized by an engine-driven fuel pump and its flow to the fuel manifold is controlled by the fuel control unit (FCU) and a starting flow control unit or a flow divider and dump valve according to engine model requirements.

The power turbine drives a propeller through a two-stage planetary reduction gearbox located at

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the front of the engine. The gearbox embodies an integral torquemeter device which is instrumented to provide an accurate indication of engine power.

The propeller reversing installation consists of a single-acting hydraulic propeller controlled by a propeller governor, combining the functions of a normal constant speed unit (CSU), a reversing valve and a power turbine (Nf) governor. A mechanical linkage between the propeller governor Beta control valve and the air bleed link enables the FCU and the propeller governor to modify engine power to maintain power turbine speed at a speed slightly less than the selected rpm when operating in the Beta control range.

2. Engine Data

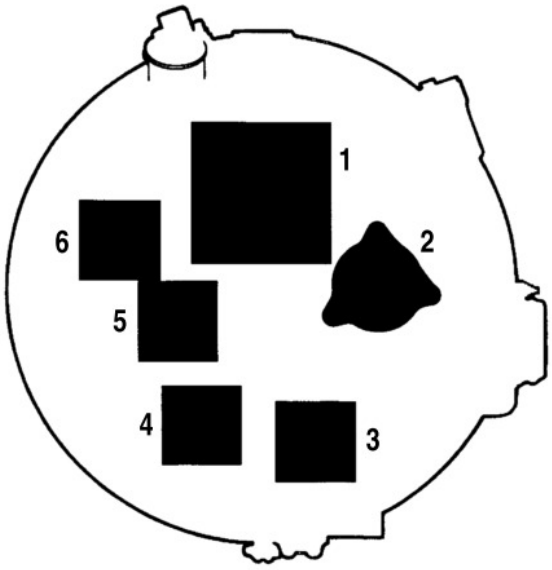
For engine specifications and leading particulars, refer to Tables 1, 3 and 5.

For the front and rear accessory drives-leading particulars, refer to Table 7.

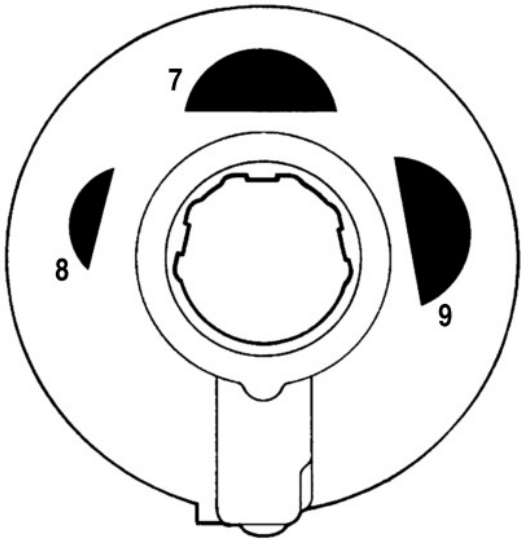
For engine fuel and lubrication system specifications and leading particulars, refer to Table 8.

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Figure 5 Engine Accessory Drives



AS VIEWED FROM REAR



AS VIEWED FROM FRONT

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Table 1 PT6A-21 Engine Specifications (Sea Level Static Output - Standard Day)

OPERATING CONDITION	ESHP	SHP	JP-4 FUEL CONSUMPTION (lb/eshp/hr) at 15°C (59°F)	JET THRUST (lb.)
Takeoff	580 (4)	550 (1)	0.630	75
Max. Continuous	580 (4)	550 (1)	0.630	75
Max. Climb (2)	580 (4)	550 (3)	0.630	75
Max. Cruise (2)	580 (4)	550 (3)	0.630	75
(1) Available to 32.8 °C (91 °F) ambient.				
(2) No certification ratings.				
(3) Available to 47.2 °C (117 °F) ambient.				
(4) Ratings are for 2205 propeller rpm.				

Table 2 PT6A-21 Engine Leading Particulars

LEADING PARTICULARS	
Engine type	Free turbine
Type of combustion chamber	Annular
Compression ratio	7.0:1
Propeller shaft rotation (looking FWD)	Clockwise
Propeller shaft configuration	Flanged
Propeller shaft gear ratio	0.0664:1
Engine diameter (combustion case approx. at room temp.)	19.0 in. (483 mm)
Engine length (approx. at room temp.)	62.0 in. (1575 mm)
Oil consumption	0.2 lb/hr (100 cc/hr) or 1 liter/10 hr
Dry weight (Incl. standard equip.)	328 lb. (148.8 kg)

Table 3 PT6A-27 Engine Specifications (Sea Level Static Output - Standard Day)

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OPERATING CONDITION	ESHP	SHP	JP-4 FUEL CONSUMPTION (lb/eshp/hr) at 15°C (59°F)	JET THRUST (lb.)
Takeoff	715 (4)	680 (1)	0.602	87.5
Max. Continuous	715 (4)	680 (1)	0.602	87.5
Max. Climb (2)	652 (4)	620 (3)	0.612	80
Max. Cruise (2)	652 (4)	620 (3)	0.612	80
(1) Available to 21.7 °C (71 °F) ambient.				
(2) No certification ratings.				
(3) Available to 20.6 °C (69 °F) ambient.				
(4) Ratings are for 2205 propeller rpm.				

Table 4 PT6A-27 Engine Leading Particulars

LEADING PARTICULARS	
Engine type	Free turbine
Type of combustion chamber	Annular
Compression ratio	7.0:1
Propeller shaft rotation (looking FWD)	Clockwise
Propeller shaft configuration	Flanged
Propeller shaft gear ratio	0.0664:1
Engine diameter (combustion case approx. at room temp.)	19.0 in. (483 mm)
Engine length (approx. at room temp.)	62.0 in. (1575 mm)
Oil consumption	0.2 lb/hr (100 cc/hr) or 1 liter/10 hr
Dry weight (Incl. standard equip.)	328 lb. (148.8 kg)

Table 5 PT6A-28 Engine Specifications (Sea Level Static Output - Standard Day)

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OPERATING CONDITION	ESHP	SHP	JP-4 FUEL CONSUMPTION (lb/eshp/hr) at 15°C (59°F)	JET THRUST (lb.)
Takeoff	715 (3)	680 (1)	0.602	87.5
Max. Continuous	715 (3)	680 (1)	0.602	87.5
Max. Climb (2)	652 (3)	620 (4)	0.612	80
Max. Cruise (2)	652 (3)	620 (5)	0.612	80
(1) Available to 21.7 °C (71 °F) ambient.				
(2) No certification ratings.				
(3) Ratings are for 2205 propeller rpm.				
(4) Available to 20.6 °C (69 °F) ambient.				
(5) Available to 33 °C (91 °F) ambient.				

Table 6 PT6A-28 Engine Leading Particulars

LEADING PARTICULARS	
Engine type	Free turbine
Type of combustion chamber	Annular
Compression ratio	7.0:1
Propeller shaft rotation (looking FWD)	Clockwise
Propeller shaft configuration	Flanged
Propeller shaft gear ratio	0.0664:1
Engine diameter (combustion case approx. at room temp.)	19.0 in. (483 mm)
Engine length (approx. at room temp.)	62.0 in. (1575 mm)
Oil consumption	0.2 lb/hr (100 cc/hr) or 1 liter/10 hr
Dry weight (Incl. standard equip.)	328 lb. (148.8 kg)

Table 7 Accessory Drives - Leading Particulars

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Drive Pad		Rotation	Ratio	Max. Torque (in.lb.)		Max. Speed (RPM)
				Continuous	Static	
1	Starter-generator (1)	CW	0.2931:1	170	1600	10,982
2	Fuel Pump/FCU (1)	CCW	0.1670:1	-	-	6,257
3	Ng Tach Generator (1)	CCW	0.1121:1	7	100	4,200
4	Vacuum Air Pump (Optional) (1)	CCW	0.1019:1	60	800	3,818
5	Optional Accessory Drive (1)	CW	0.3208:1	135	800	12,027
6	Optional Accessory Drive (Hydraulic Pump) (1)	CCW	0.2041:1	150	800	7,643
7	Propeller Governor (2)	CW	0.1273:1 (PT6A-21) 0.1264 (PT6A-27/-28)	50	850	-
8	Nf Tach Generator (2)	CW	0.1273:1 (PT6A-21) 0.1264 (PT6A-27/-28)	7	100	-
9	Propeller OverspeedGovernor (2)	CW	0.1273:1 (PT6A-21) 0.1264 (PT6A-27/-28)	50	850	-
NOTE: 1. Rear Accessory Drives: 100% is 37,468 rpm Ng.						
NOTE: 2. Front Accessory Drives: 100% is 33,235 rpm Nf.						

Table 8 Fuel and Lubrication System Specifications

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Nomenclature	Specification
Fuel Specification	Ref. P&WC SB1244 (Ref. NOTE 1)
Oil Specification	Ref. P&WC SB1001 (Ref. NOTE 1)
Oil Tank - Total Capacity	2.8 US gallons (2.33 Imp. gallons, 10.6 liters)
Oil Tank - Expansion Space	0.7 US gallons (0.58 Imp. gallons, 2.65 liters)
Oil Tank - Usable Quantity	1.5 US gallons (1.25 Imp. gallons, 5.68 liters)
NOTE: 1. Ref. Service Bulletin List for further details.	

3. Approved Service Bulletins

The following Service Bulletins contain data recommended by Pratt & Whitney Canada and approved by the Canadian Minister of Transport.

SB1001: Approved Listing of Synthetic Oils.

SB1002: Rotor Components - Service Life.

SB1803: Operating Time Between Overhauls and Hot Section Inspection Frequency

SB1244: Requirements and Approved Listing of Engine Fuel And Additives.

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